

- 10 Two satellites P and Q orbit the Earth due to the gravitational field of the Earth. P and Q are at distances R and $3R$ from the Earth's surface respectively, where R is the radius of the earth. The speed of P is v .

What is the speed of Q?

A $\sqrt{\frac{1}{3}} v$

B $\sqrt{\frac{1}{2}} v$

C $\sqrt{2} v$

D $\sqrt{3} v$

Ans: B

Gravitational force provides for centripetal force.

$$F_G = F_C$$

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{GM}{r}}$$

$$v_P = \sqrt{\frac{GM}{2R}}$$

$$v_Q = \sqrt{\frac{GM}{4R}} = \left(\frac{1}{\sqrt{2}}\right) \sqrt{\frac{GM}{2R}} = \frac{1}{\sqrt{2}} v$$

Take note that the satellites are R and $3R$ from the surface of Earth, not the centre of the Earth.

- 11 Mars has a diameter that is approximately 0.5 times the diameter of the Earth, and the mass of